AI-Based Battery Lifecycle Assessment and Second-Life Management Using Digital Battery Passports

Week 5: Remaining useful life prediction

Mon 30 Nov - Fri 4 Dec 2026 · Sprint 5 · Overall progress 55%

Student name		**Roll no.**	
Faculty in charge		**Date submitted**	Fri 4 Dec 2026

On track **This week in one line:** Quantile Gradient Boosting predicts cycles left to 80% SOH within 8.2 cycles (NASA) and 46.6 cycles (CALCE); calibrated bands now cover 76-84% at an 80% target.

1. Goal for the week

Predict how many cycles remain before 80% SOH (end of first life) and 60% SOH (end of second life), with calibrated uncertainty.

2. Day-by-day work

Day	Work done
Mon 30 Nov	Ground-truth end-of-life from smoothed SOH; causal snapshots so the predictor only sees cycles 1..k.
Tue 1 Dec	Baselines: linear trend and an empirical fade model with an accelerating 'knee' term.
Wed 2 Dec	Quantile Gradient Boosting (10/50/90%) on SOH estimate, fade rate, fade acceleration, charge signals, resistance.
Thu 3 Dec	Raw bands covered only 22-33%; implemented conformalised quantile regression (CQR).
Fri 4 Dec	Evaluation by life stage, figure, 5 new tests (21 total), this report.

3. Completed deliverables

- **batterylife/models/rul.py**.
- **results/rul_metrics.json**, rul_predictions.csv.
- tests/test_rul.py.

4. Results

Dataset	Mean life to 80%	Linear trend	Fade model	Quantile GBR	Band coverage
NASA	72 cycles	78.9	55.9	**8.2**	75.9%
CALCE	578 cycles	388.8	145.4	**46.6**	78.6%
CALCE to 60%	790 cycles	499.4	352.8	**55.4**	84.0%

Mean absolute error in cycles on unseen cells.

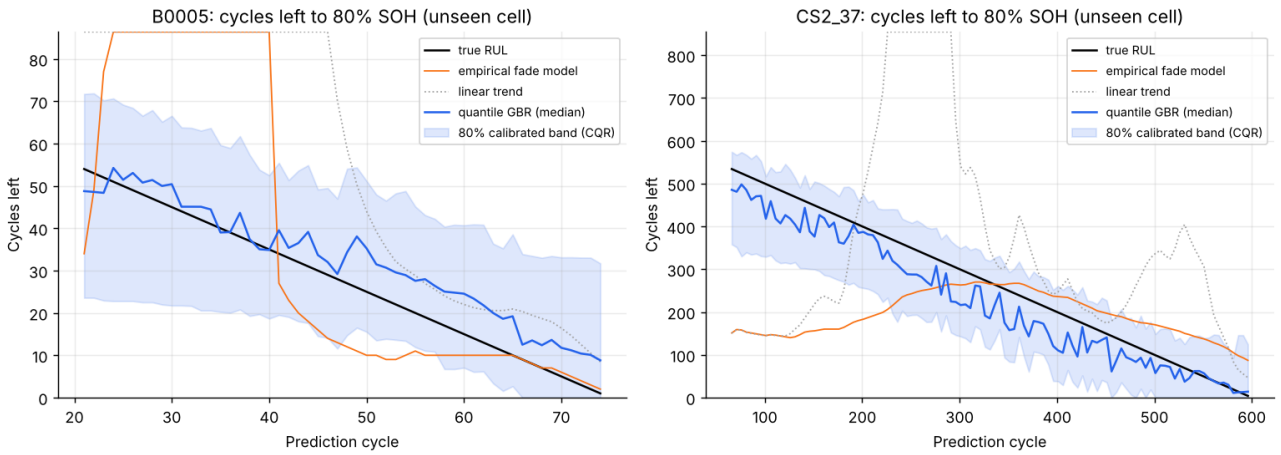


Figure 5. Cycles left to 80% SOH on two unseen cells: truth, baselines and the calibrated band.

5. Problems faced and how they were solved

Problem	Resolution
Curve extrapolation cannot see the CALCE knee before it happens.	The learned model anticipates it from the other cells.
NASA capacity regeneration flattens fitted trends.	Same: data-driven model; baselines kept for comparison.
Uncalibrated bands were dangerously narrow.	CQR brings coverage close to the 80% target.

6. Plan for next week

- Turn SOH and RUL into second-life grades and application matches.
- Quantify CO2 and value of re-use.

Faculty remarks

Faculty signature: _____ Student signature: _____